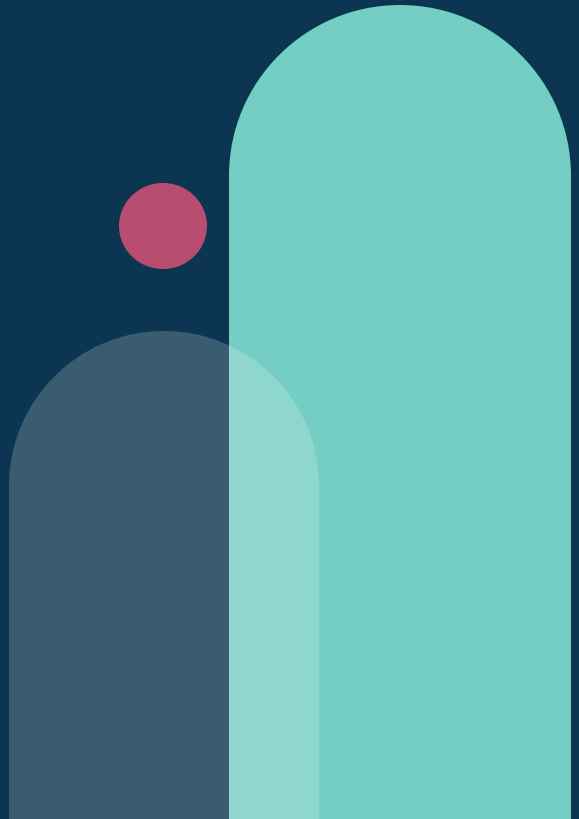
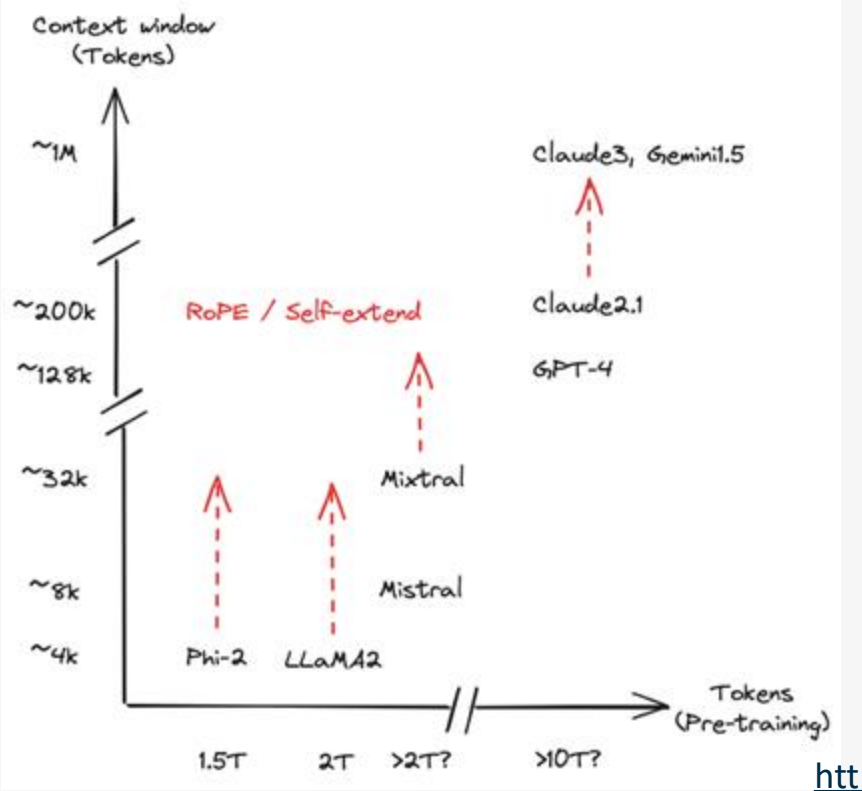
Retrieval-Augmented Generation (RAG)

01

Overview



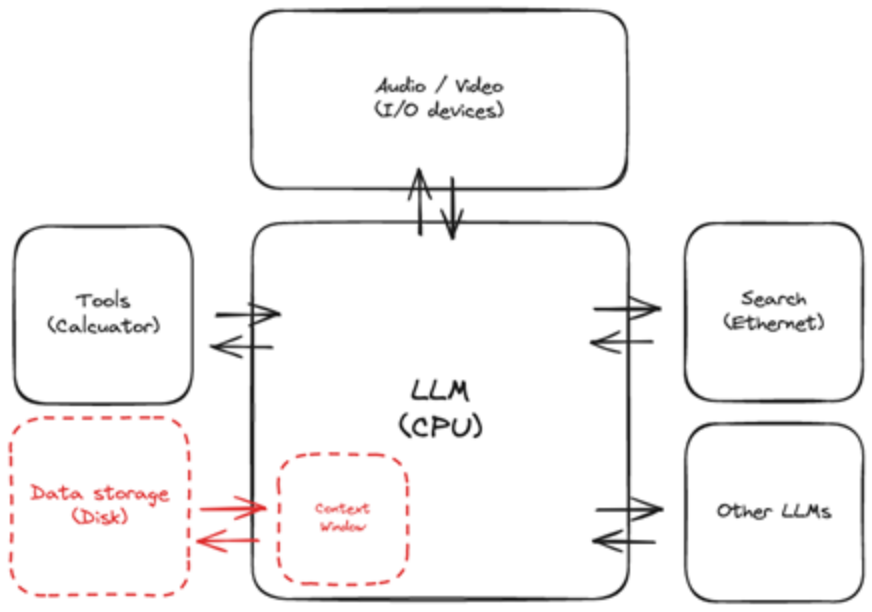
> 95% of the world's data is "private", but we can "feed it" to LLMs



https://huggingface.co/blog/mixtral

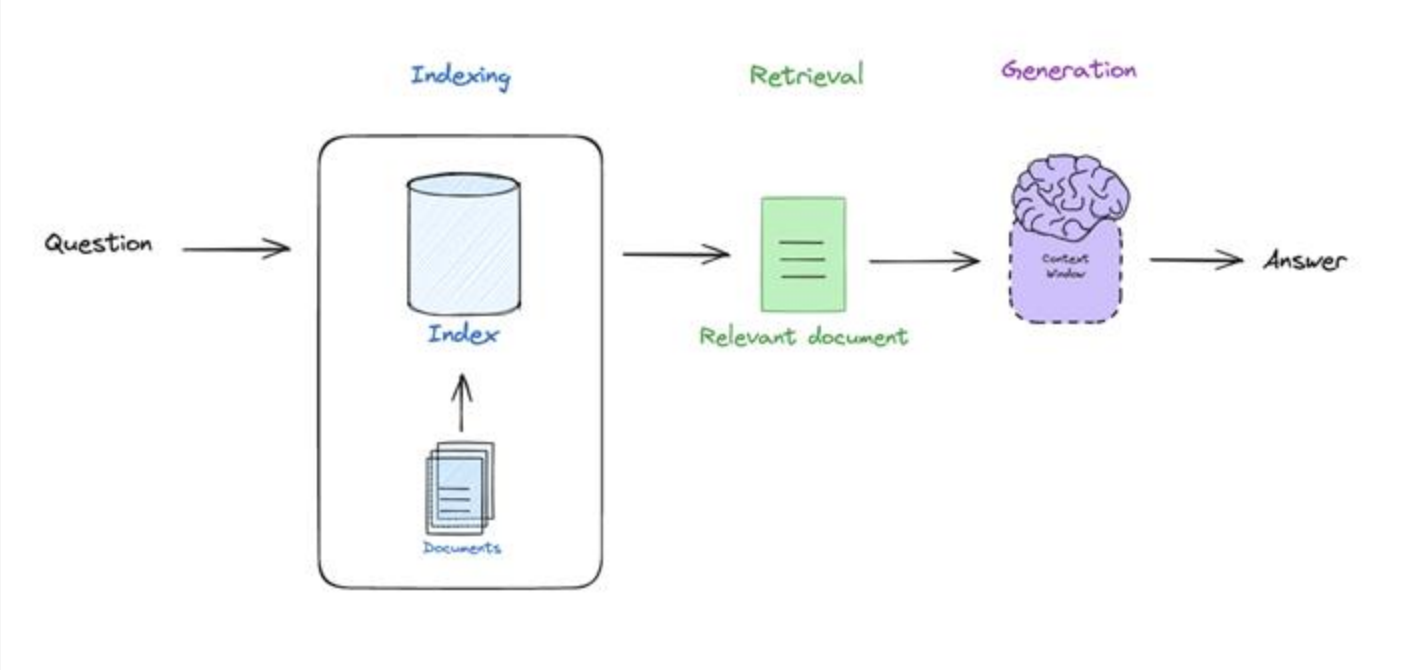
https://x.com/RihardJarc/status/1778082161595208124

Connecting LLMs to external data is a central need

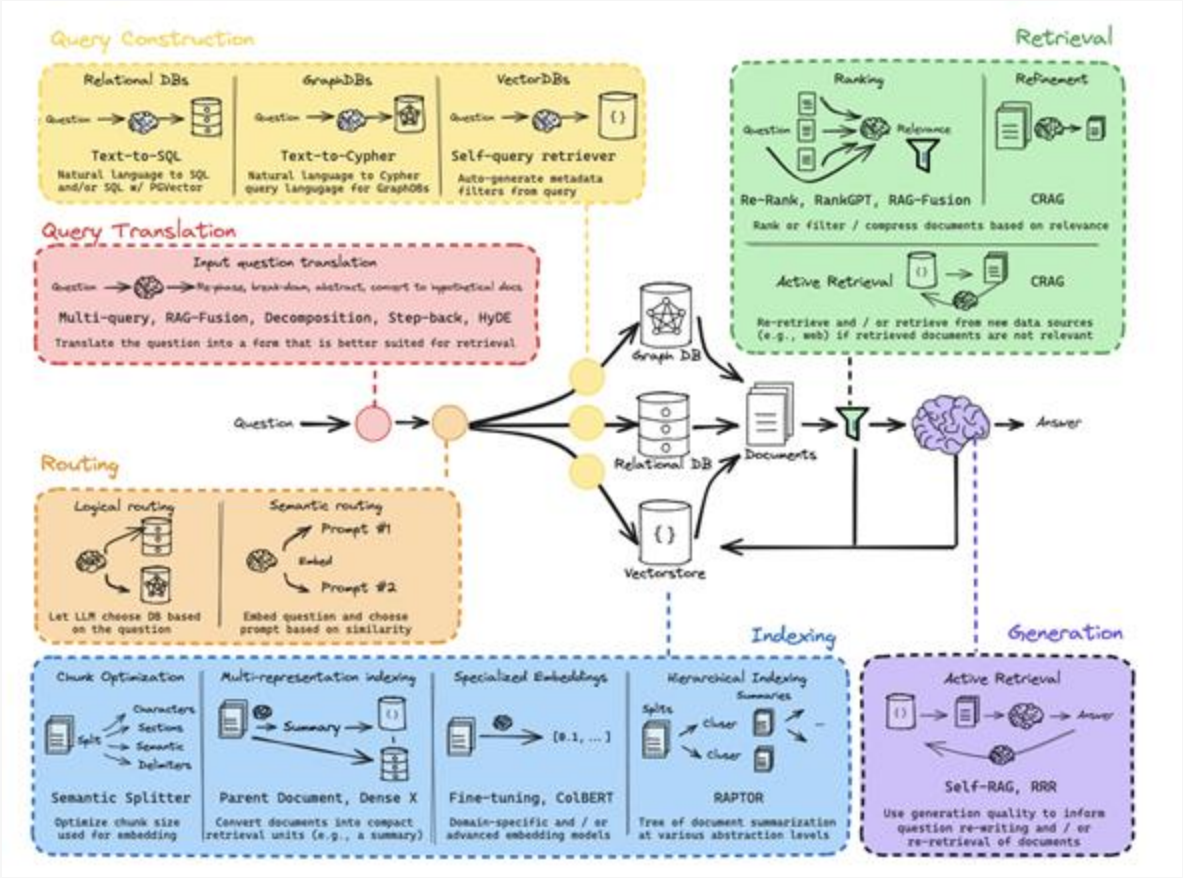


https://x.com/karpathy/status/1707437820045062561?s=20

Retrieval **A**ugmented **G**eneration



We’ll start from scratch, and build up to state of the art in RAG



Outline

Basics

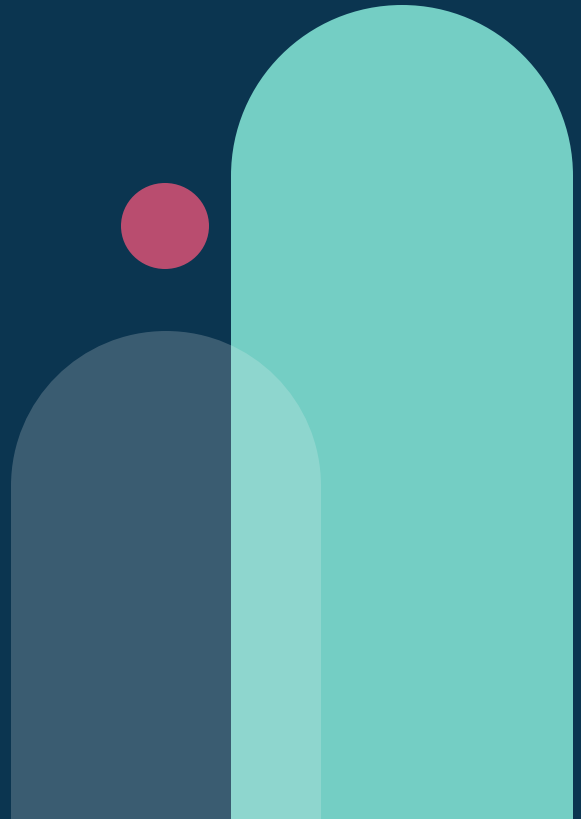
- Indexing
- Retrieval
- Generation

Advanced

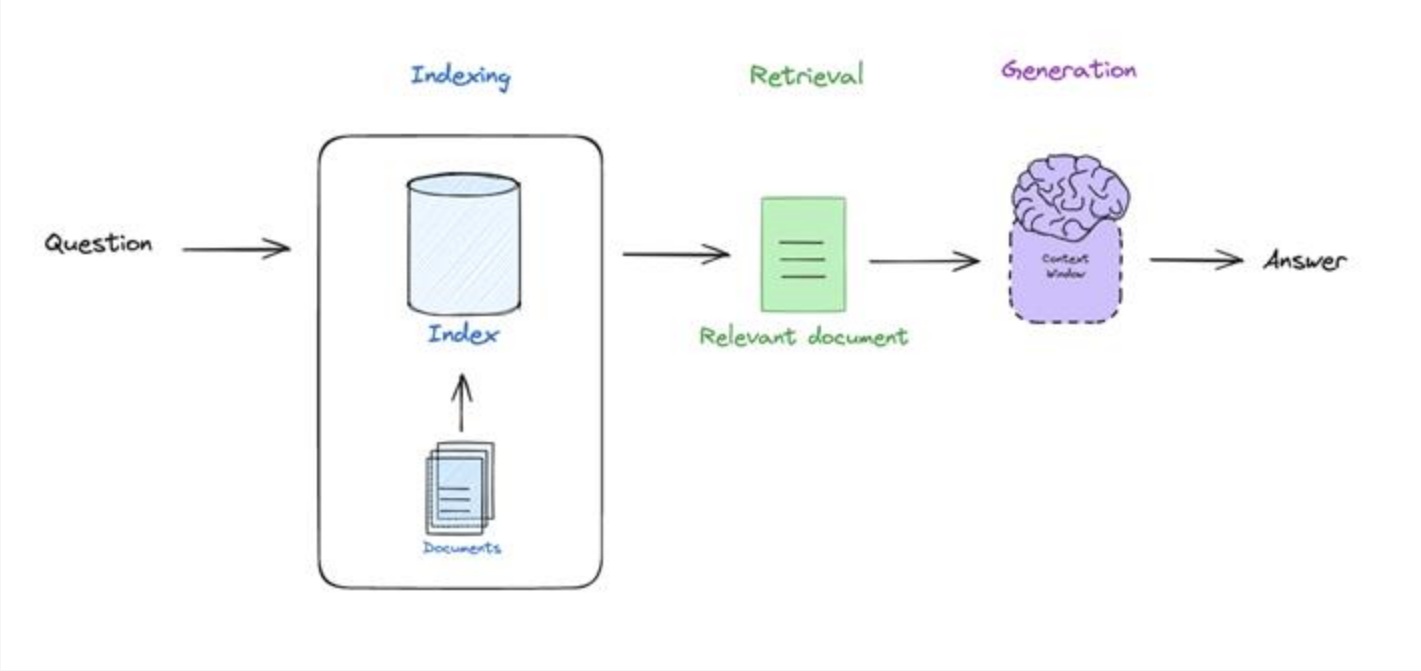
- Query transformations
- Routing
- Query construction
- Indexing
- Retrieval
- Generation

02

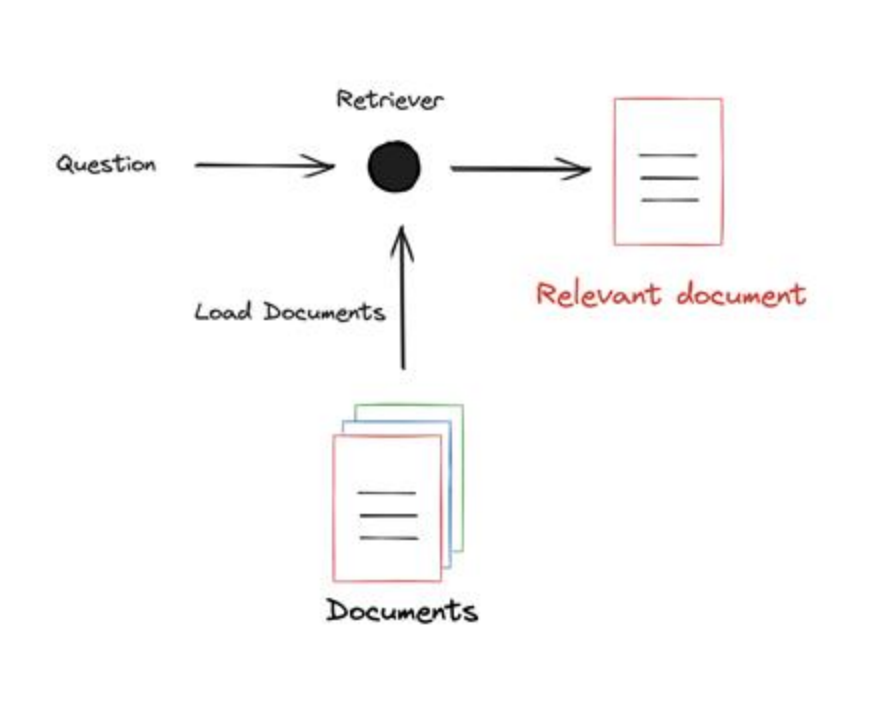
Indexing



RAG motivation

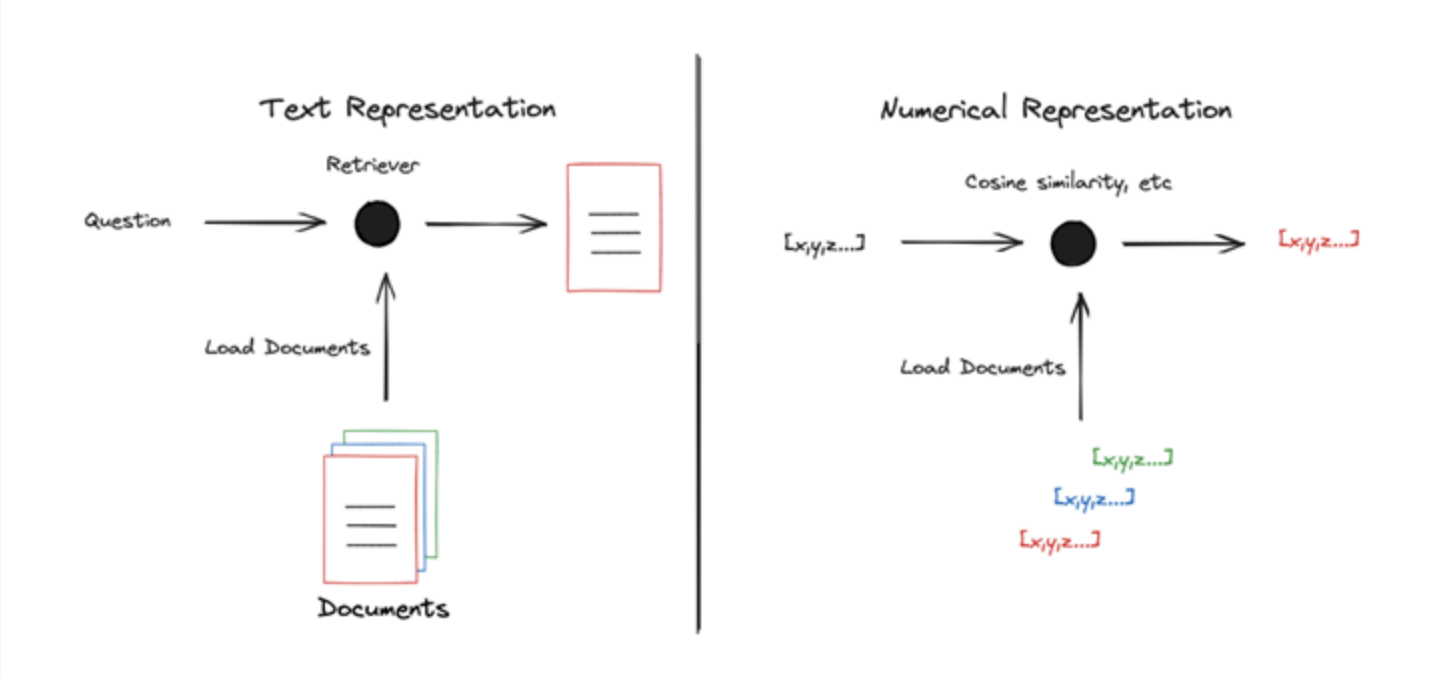


Document loading

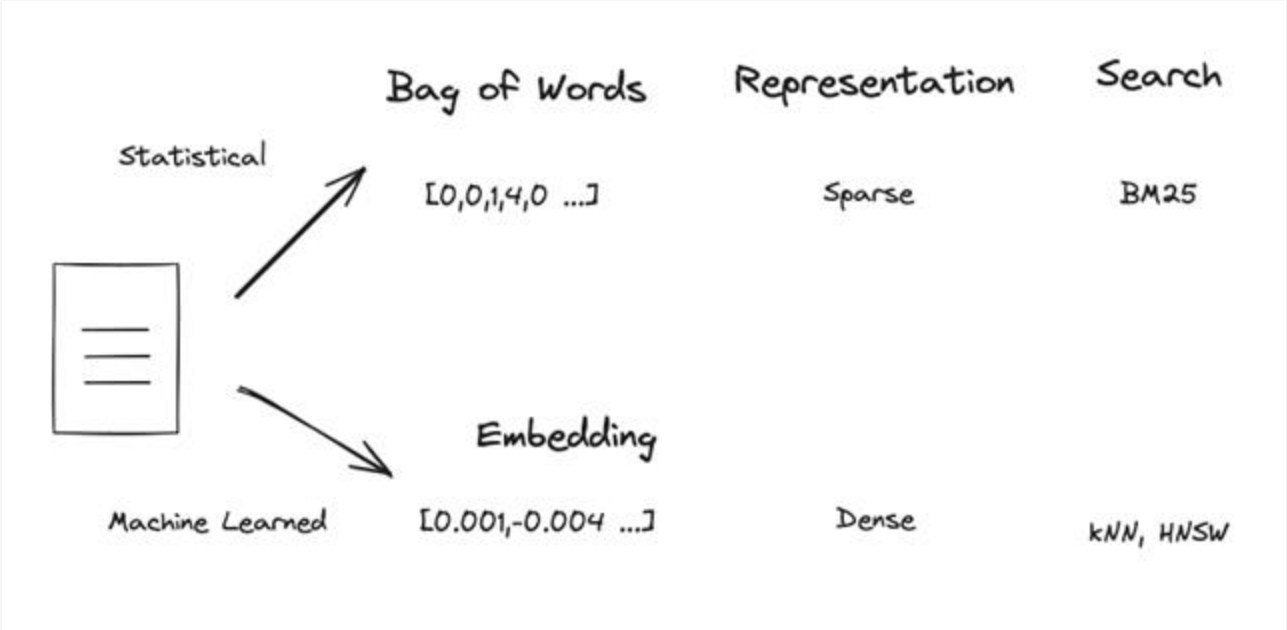


https://integrations.langchain.com/

Numerical representation for search

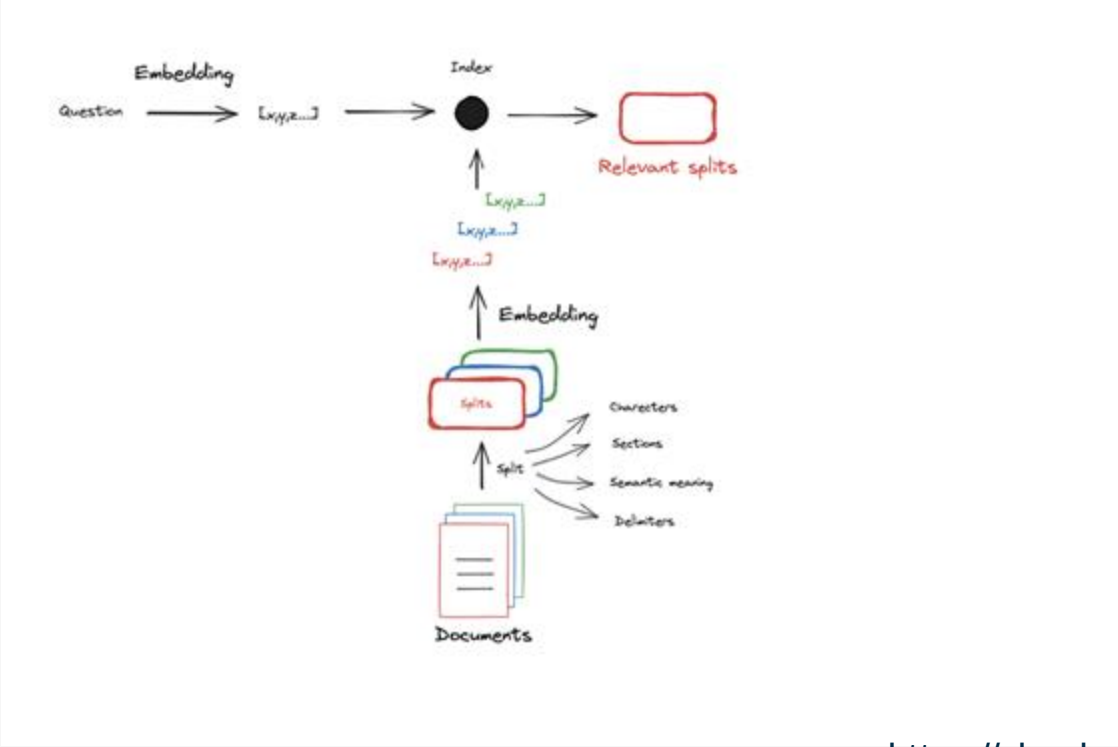


Statistical and machine learned representations



https://simonwillison.net/2023/Oct/23/embeddings/
https://www.pinecone.io/learn/series/nlp/dense-vector-embeddings-nlp/
https://cameronrwolfe.substack.com/p/the-basics-of-ai-powered-vector-search?utm_source=profile&utm_medium=reader2

Loading, splitting, and embedding

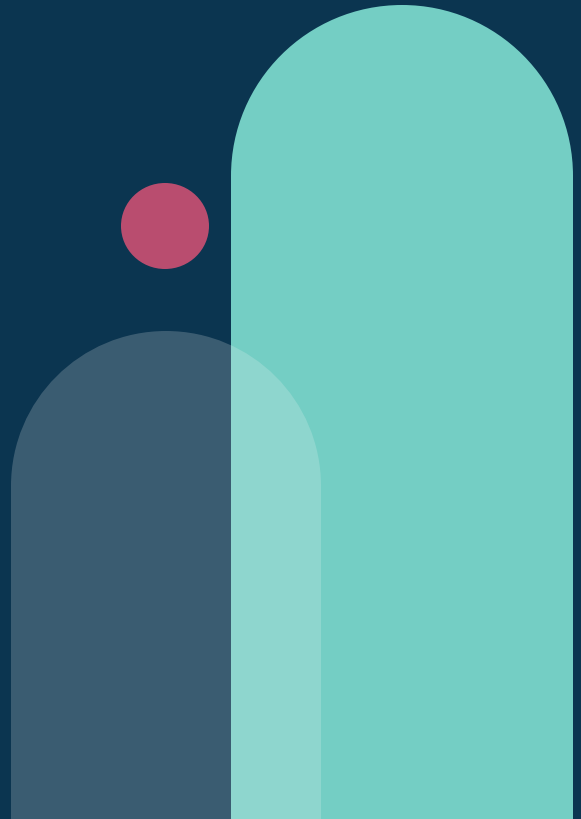


https://chunkviz.up.railway.app/

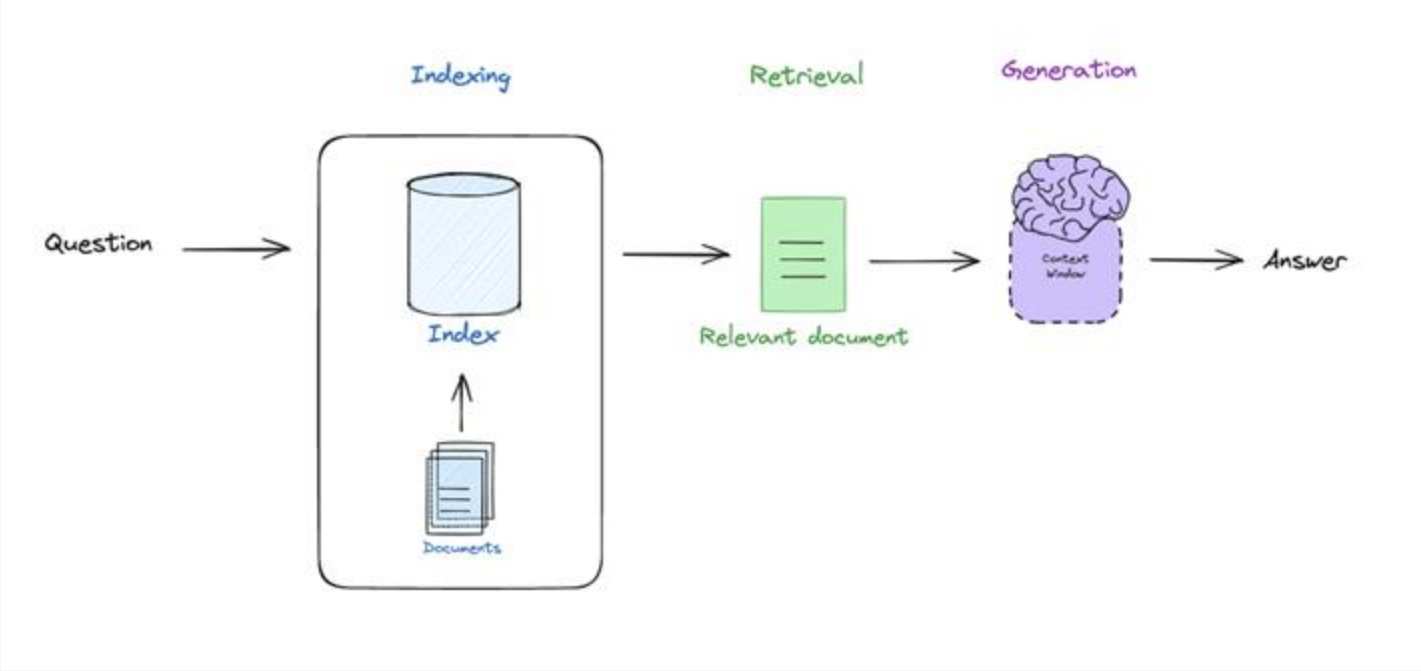
https://www.youtube.com/watch?v=8OJC21T2SL4

03

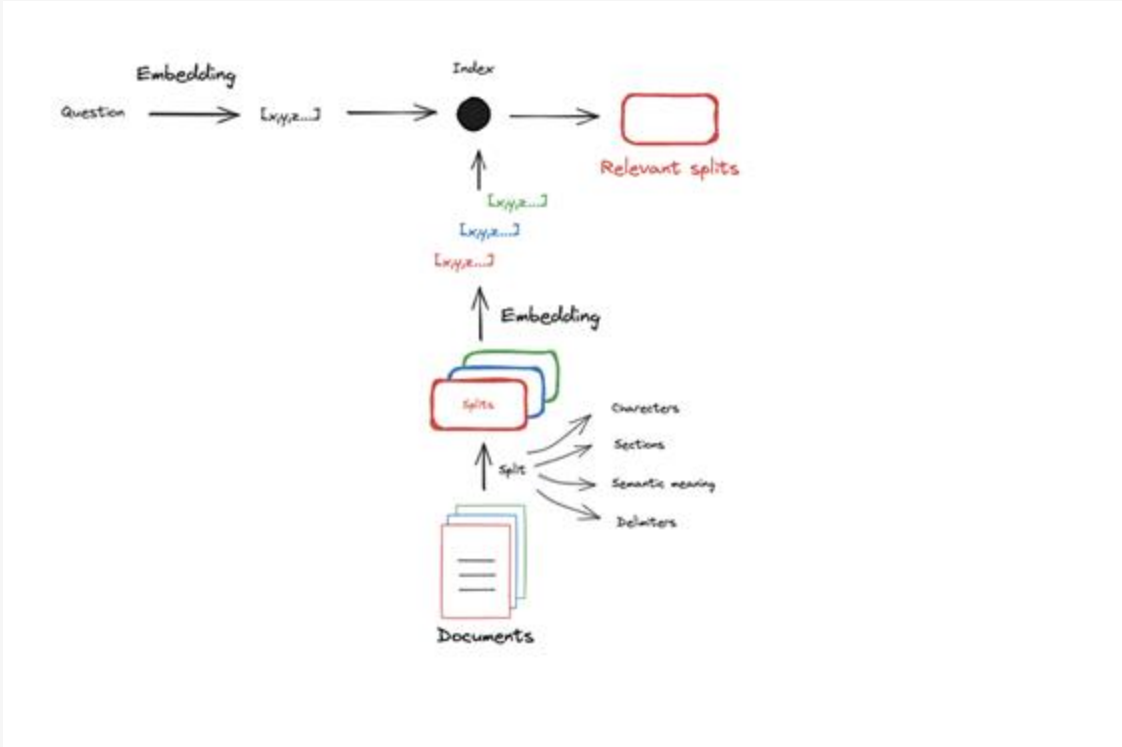
Retrieval



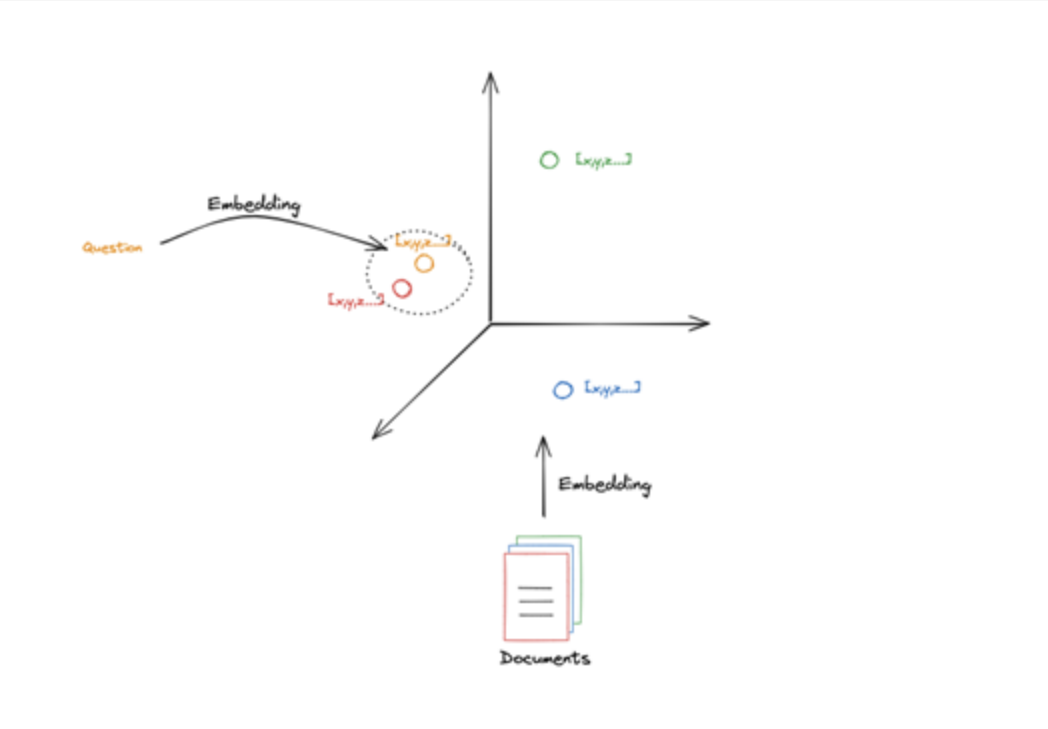
RAG motivation



Index makes documents easy to retrieve

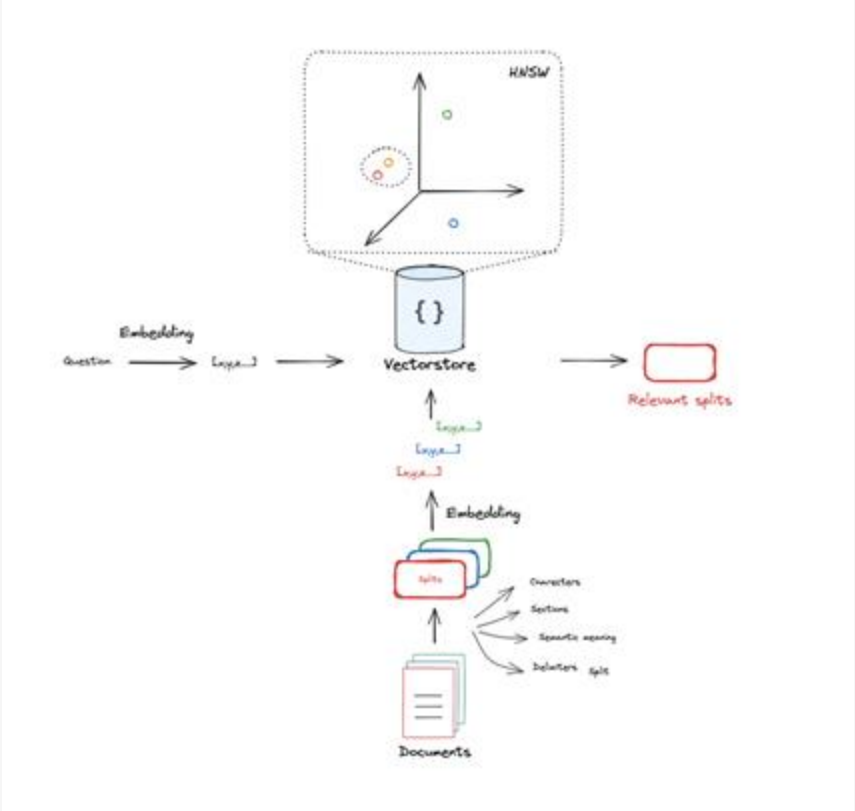


Retrieval powered via similarity search



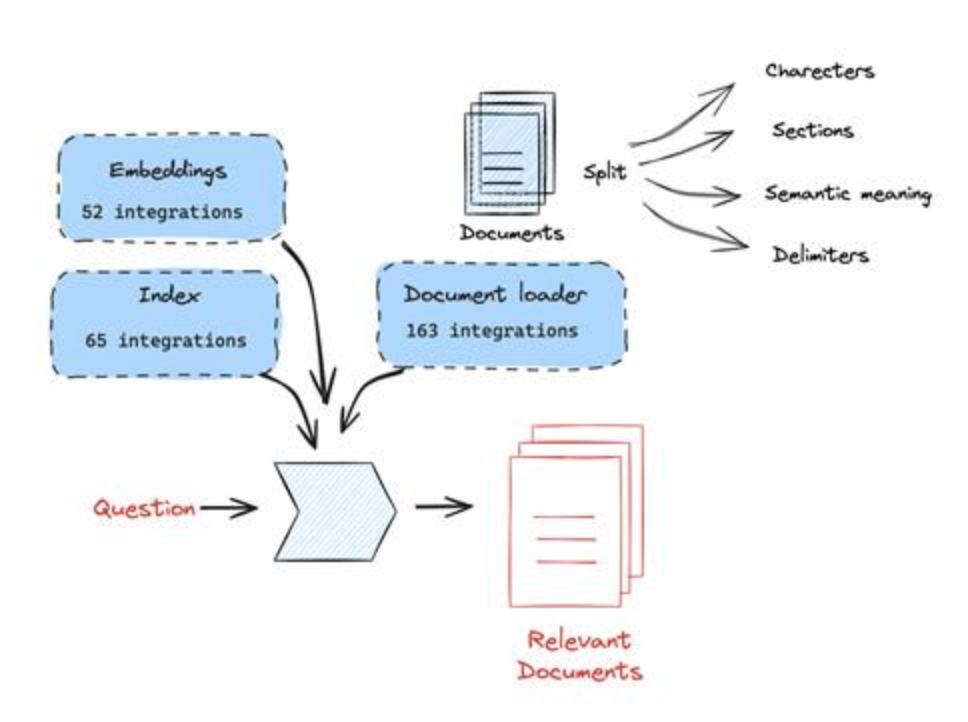
https://simonwillison.net/2023/Oct/23/embeddings/

Vectorstores implement this for you



https://www.pinecone.io/learn/series/faiss/hnsw/

LangChain has many integrations to support this

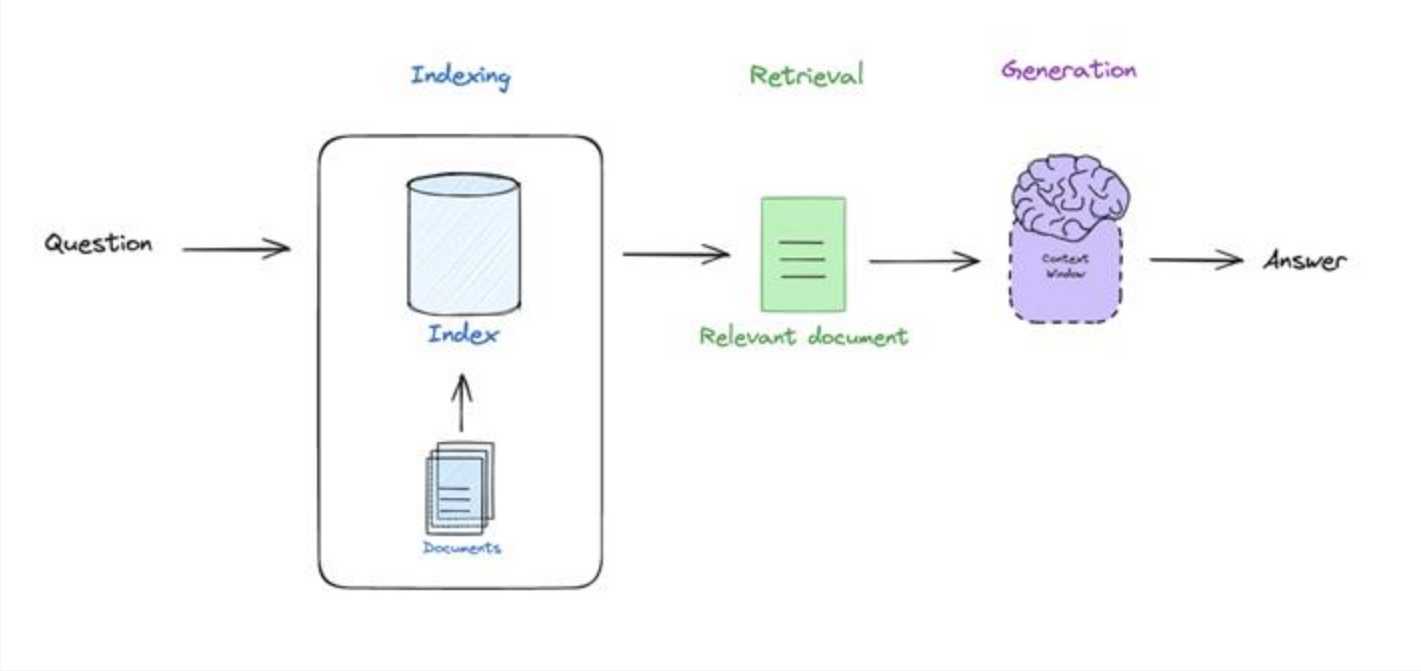


https://integrations.langchain.com/
https://vdbs.superlinked.com/

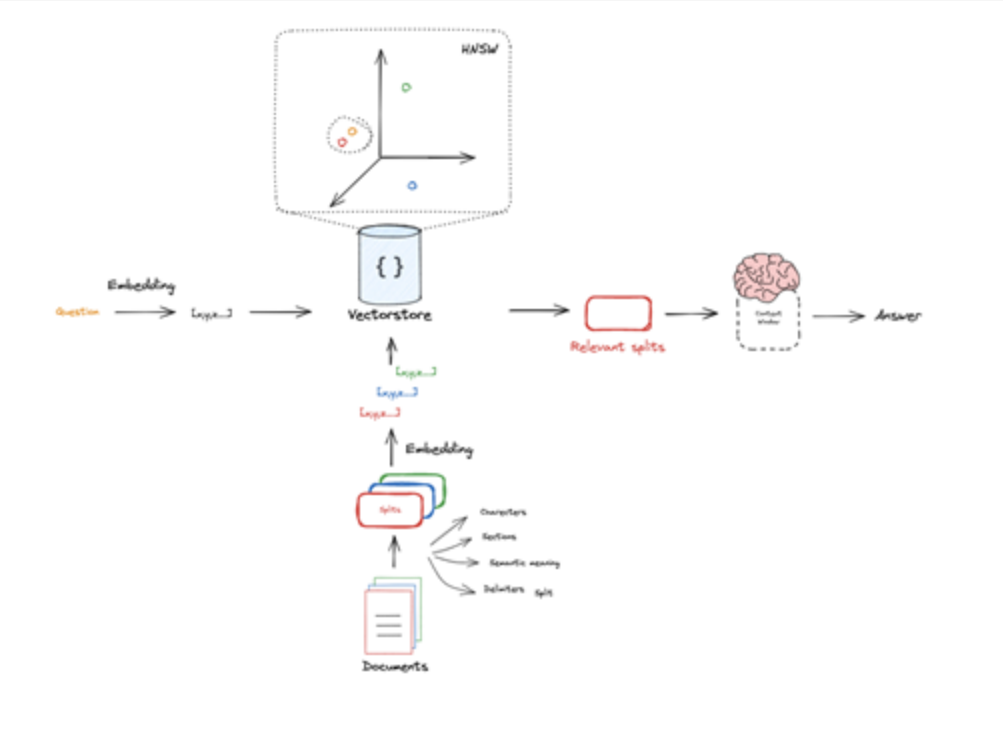
04

Generation

RAG motivation

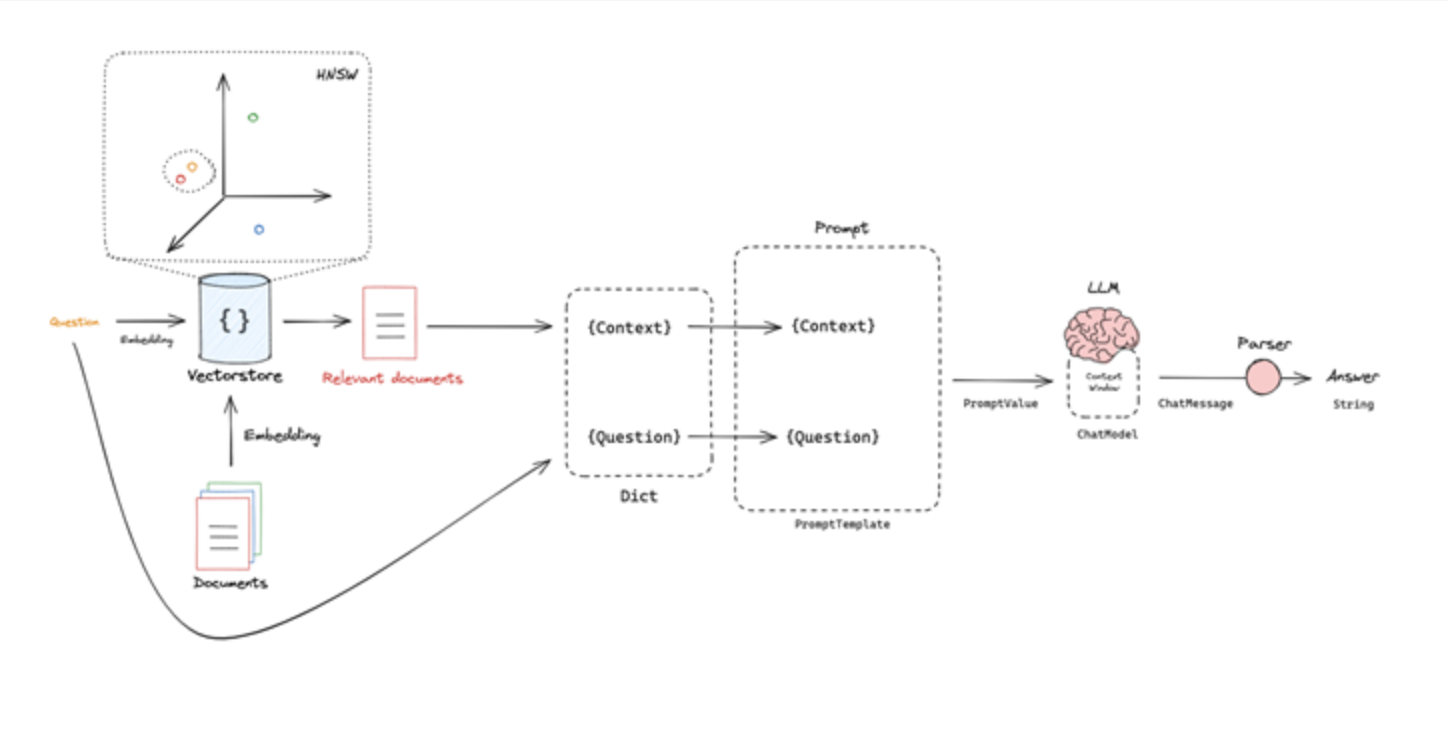


Adding docs to context window



https://python.langchain.com/docs/use_cases/question_answering/

Connecting retrieval with LLMs via prompt



https://python.langchain.com/docs/expression_language/get_started

https://smith.langchain.com/hub/rlm/rag-prompt?organizationId=1fa8b1f4-fcb9-4072-9aa9-983e35ad61b8

Thanks!

Do you have any questions?

